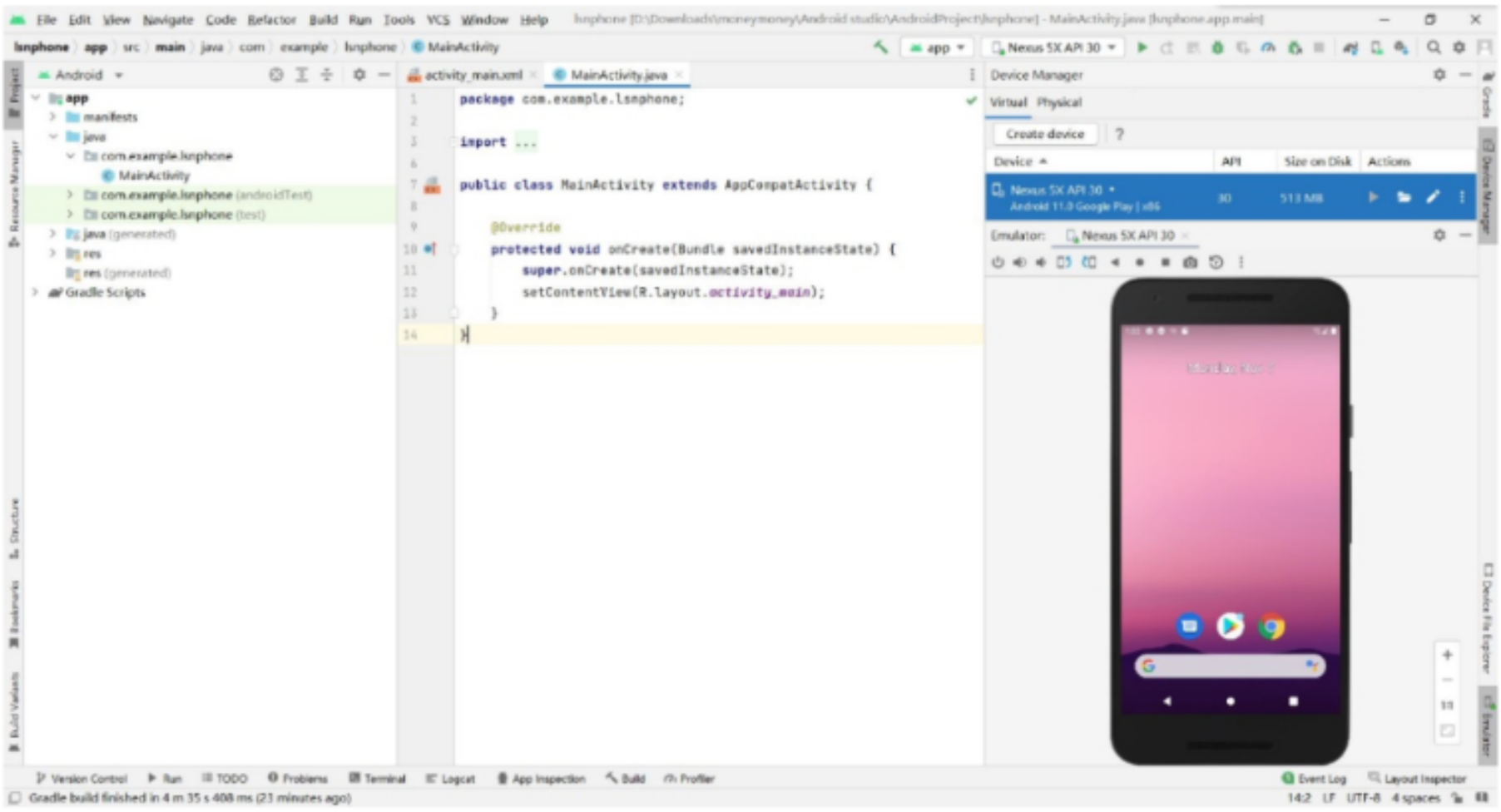
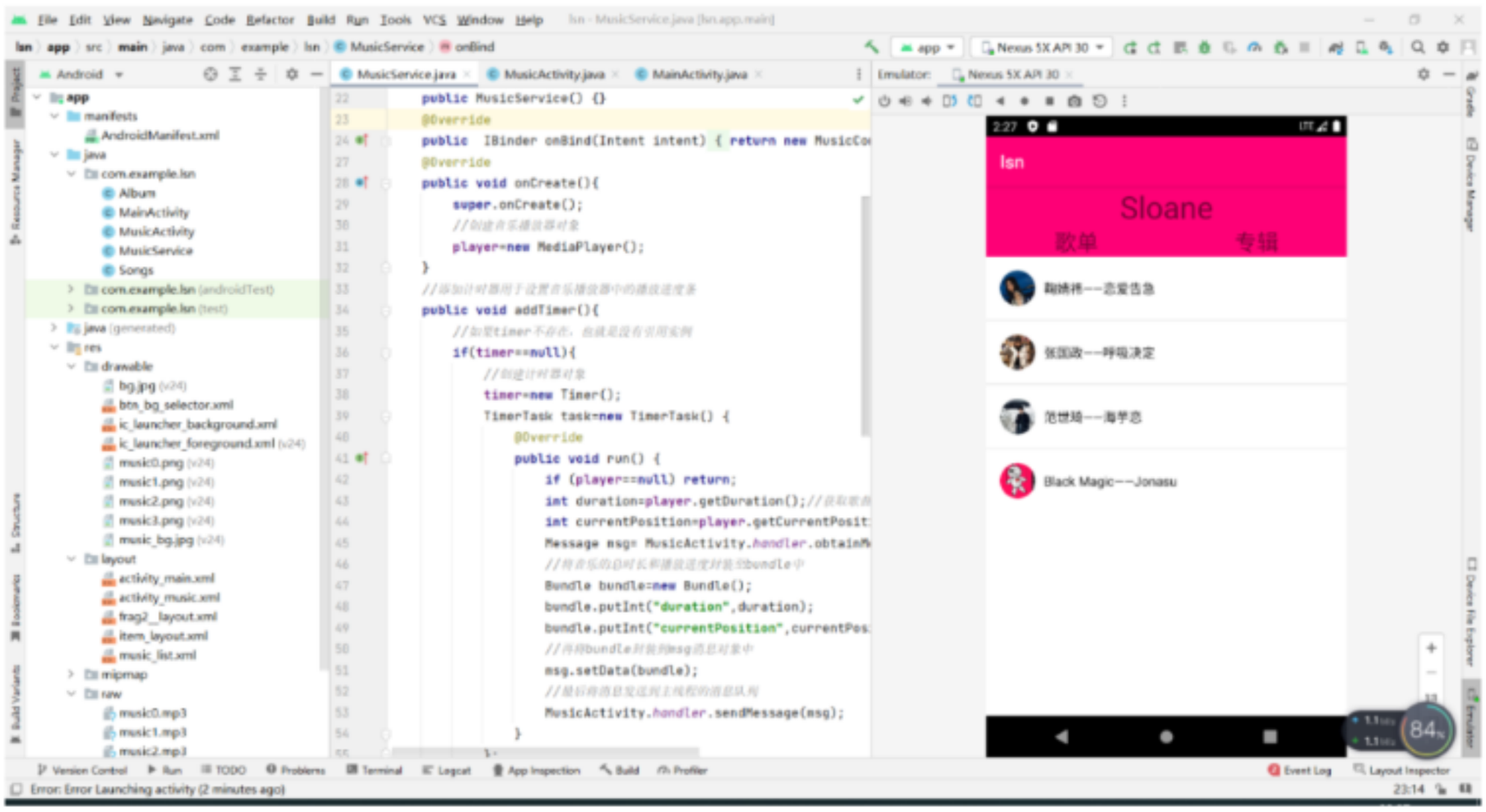
第一章作业：

P22 1. 下载最新版本的 Android Studio，安装后如果需要更新 Gradle，则将其更新到最新版本。

P22 2. 创建一个任意类型的手机 App 工程，分别使用虚拟设备和物理设备运行。





第二章作业：

P73　2.2　实现多选题的界面效果。

代码：

```
public class MainActivity extends AppCompatActivity implements
CompoundButton.OnCheckedChangeListener {
      private final ArrayList<String> mHobbies = new ArrayList<>();
         protected void onCreate(Bundle savedInstanceState) {
            super.onCreate(savedInstanceState);
            setContentView(R.layout.activity_main);
            final TextView resultTextView = findViewById(R.id.textView);
            CheckBox checkBox1 = findViewById(R.id.checkBox5);
            CheckBox checkBox2 = findViewById(R.id.checkBox4);
            CheckBox checkBox3 = findViewById(R.id.checkBox3);
            CheckBox checkBox4 = findViewById(R.id.checkBox6);
            Button submitButton = findViewById(R.id.button);
          checkBox1.setOnCheckedChangeListener(this); //CheckBox 设置监听器
            checkBox2.setOnCheckedChangeListener(this);
            checkBox3.setOnCheckedChangeListener(this);
            checkBox4.setOnCheckedChangeListener(this);
            submitButton.setOnClickListener(v -> {//Button 设置监听器
                  StringBuilder sb = new StringBuilder();
                  for (int i = 0; i < mHobbies.size(); i++) {
                        if (i == (mHobbies.size() - 1)) {
                              sb.append(mHobbies.get(i));
                        } else {
                              sb.append(mHobbies.get(i)).append("、");
                        }}
                  if (sb.length() == 0) { //显示选择结果
                        resultTextView.setText("您还没有进行选择了！");
                  } else {
                        resultTextView.setText("您选择了：" + sb + "。");
                  } }); }
            public void onCheckedChanged(CompoundButton compoundButton,
boolean isChecked) {
            if (isChecked) {
                  mHobbies.add(compoundButton.getText().toString().trim());
            } else {
mHobbies.remove(compoundButton.getText().toString().trim());
            }}}
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
      xmlns:app="http://schemas.android.com/apk/res-auto"
      xmlns:tools="http://schemas.android.com/tools"
```

```
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<CheckBox
    android:id="@+id/checkBox5"
    android:layout_width="352dp"
    android:layout_height="78dp"
    android:rotationX="-1"
    android:textSize="30dp"
    android:text="钓鱼"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.493"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.104" />
<CheckBox
    android:id="@+id/checkBox4"
    android:layout_width="352dp"
    android:layout_height="78dp"
    android:rotationX="-1"
    android:text="游泳"
    android:textSize="30dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.491"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.269" />
<CheckBox
    android:id="@+id/checkBox3"
    android:layout_width="352dp"
    android:layout_height="78dp"
    android:rotationX="-1"
    android:textSize="30dp"
    android:text="喝水"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.491"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.415" />
<CheckBox
    android:id="@+id/checkBox6"
```

```
        android:layout_width="352dp"
        android:layout_height="78dp"
        android:rotationX="-1"
        android:textSize="30dp"
        android:text="篮球"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.491"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.572" />
    <TextView
        android:id="@+id/textView"
        android:layout_width="362dp"
        android:layout_height="43dp"
        android:gravity="center"
        android:text="你的爱好是什么?"
        android:textSize="20dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.023" />
    <Button
        android:id="@+id/button"
        android:layout_width="164dp"
        android:layout_height="66dp"
        android:layout_marginTop="48dp"
        android:layout_marginBottom="183dp"
        android:text="确定"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/checkBox6"
        app:layout_constraintVertical_bias="0.0" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

运行结果：

p73　2.3　实现选择出生日期的界面效果。

代码：

```
public class MainActivity extends AppCompatActivity {
        protected void onCreate(Bundle savedInstanceState) {
          super.onCreate(savedInstanceState);
          setContentView(R.layout.activity_main);
          final DatePicker datePicker = findViewById(R.id.date_picker);
          datePicker.updateDate(2021, 0, 1);
          datePicker.setOnDateChangedListener((view, year, monthOfYear,
dayOfMonth) -> {
              Calendar calendar = Calendar.getInstance();
              calendar.set(year, monthOfYear, dayOfMonth);
              @SuppressLint("SimpleDateFormat") SimpleDateFormat
simpleDateFormat = new SimpleDateFormat("yyyy 年 MM 月 dd 日");
              Toast.makeText(getApplicationContext(), "当前选择时间是：" +
simpleDateFormat.format(calendar.getTime()), Toast.LENGTH_SHORT).show();
          });
          Button button1 = findViewById(R.id.button1);
          button1.setOnClickListener(v -> {
              datePicker.updateDate(2020, 0, 1);
              Toast.makeText(getApplicationContext(), "已经恢复到默认时间！",
Toast.LENGTH_LONG).show();
          });
          Button button2 = findViewById(R.id.button2);
          button2.setOnClickListener(v -> {
              Calendar calendar = Calendar.getInstance();
              calendar.set(datePicker.getYear(), datePicker.getMonth(),
datePicker.getDayOfMonth());
              @SuppressLint("SimpleDateFormat") SimpleDateFormat
simpleDateFormat = new SimpleDateFormat("yyyy 年 MM 月 dd 日");
              Toast.makeText(getApplicationContext(), "你的出生日期是：" +
simpleDateFormat.format(calendar.getTime()), Toast.LENGTH_SHORT).show();
          }); }
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <TextView
        android:id="@+id/textView3"
        android:layout_width="425dp"
        android:layout_height="41dp"
        android:gravity="center"
        android:text="请选择你的出生日期：" />
```

```
<DatePicker
        android:id="@+id/date_picker"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:calendarViewShown="false"
        android:datePickerMode="spinner"
        android:startYear="2020" />
<Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="重置" />
    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="确定" />
</LinearLayout>
```

运行结果：


12:56
LTE
chushengriqi
请选择你的出生日期:
Jan
29
Feb
01
2020
Mar
02
2021
重置
确定
当前选择时间是：2020年02月01日

第三章

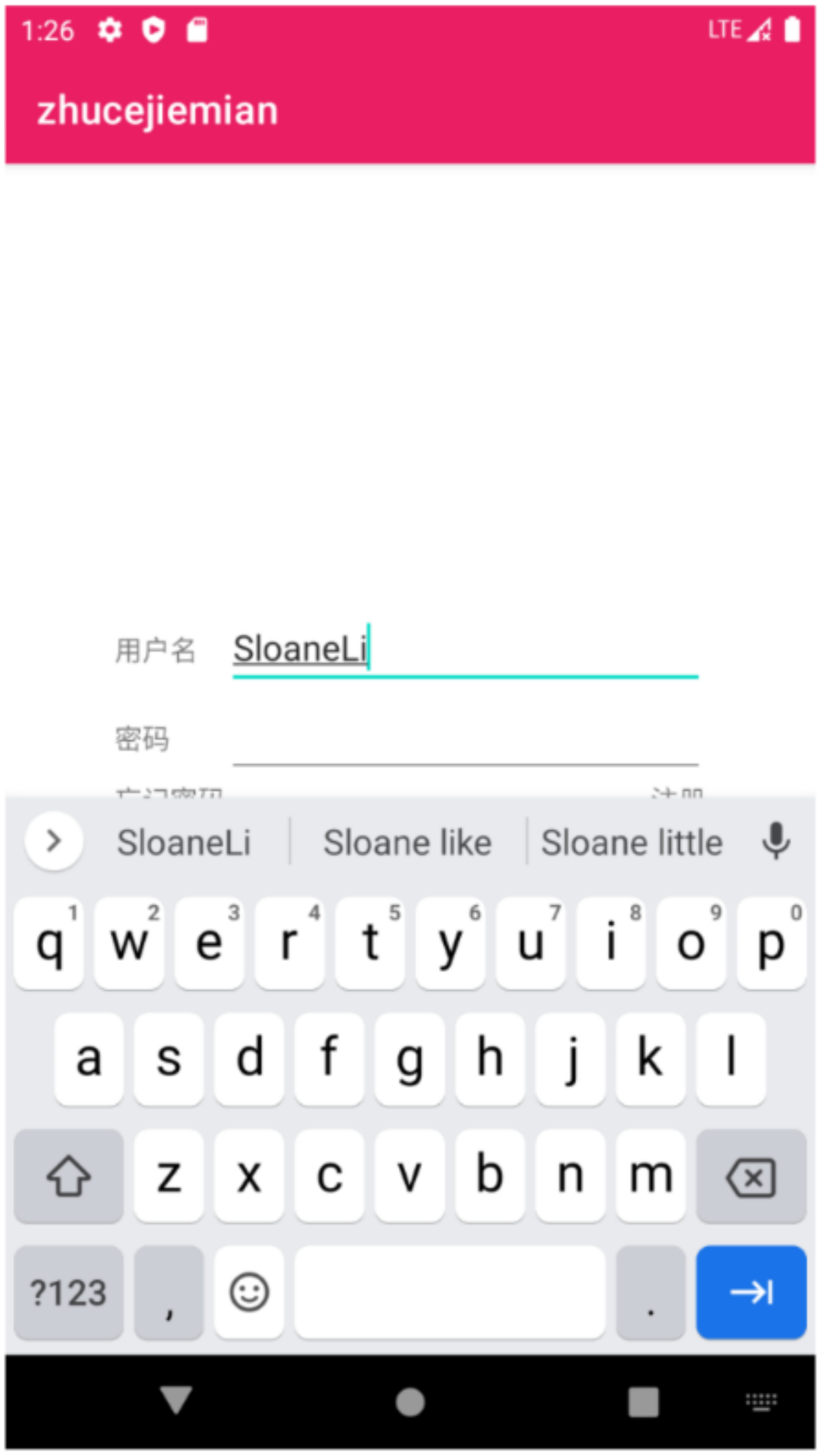
3.1　实现注册界面的布局效果，至少包括用户名、密码、忘记密码、登录等控件。

代码：

```
public class MainActivity extends AppCompatActivity {
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main); }}
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="300dp"
android:layout_height="wrap_content"
android:layout_gravity="center"
android:collapseColumns="2"
android:stretchColumns="1">
    <TableRow><TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="用户名" />
     <EditText
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:ems="10"
            android:inputType="textPersonName" />
</TableRow> <TableRow>
    <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="密码" />
    <EditText
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:ems="10"
            android:inputType="textPassword" />
    <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="忘记密码" />
</TableRow> <TableRow>
    <TextView
            android:id="@+id/textView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="忘记密码" />
    <TextView
            android:id="@+id/textView4"
```

```
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_column="1"
            android:gravity="end"
            android:text="注册" />
    </TableRow>
    <TableRow android:orientation="vertical">
        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_span="2"
            android:text="登录" />
    </TableRow></TableLayout>
```

运行结果：


1:26
LTE
zhucejiemian
用户名
SloaneLi
密码
SloaneLi
Sloane like
Sloane little
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123
,
.

第四章：

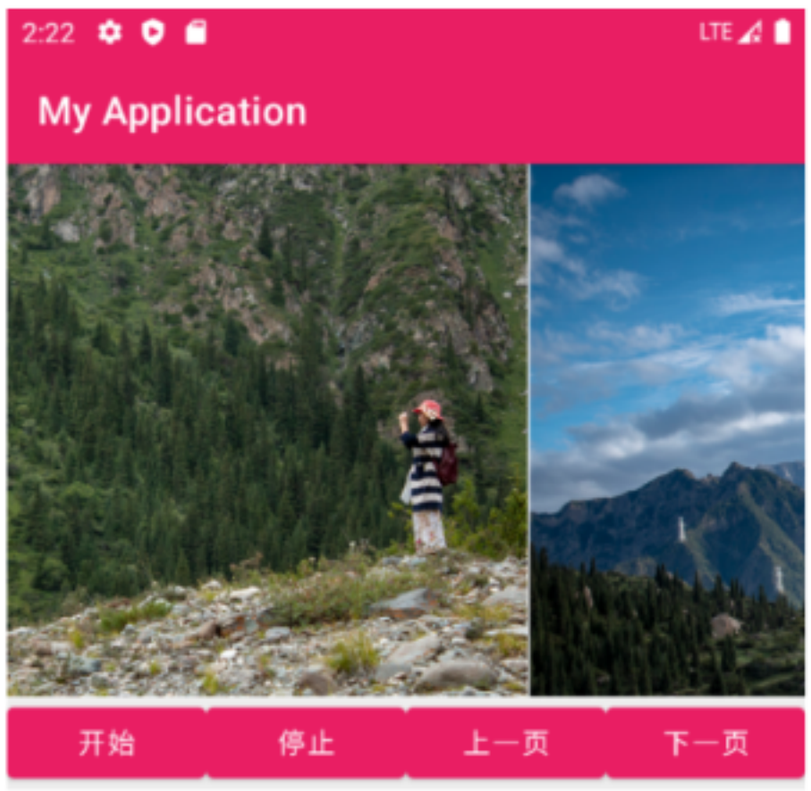
p122 4.3 实现京东首页的轮播广告效果，至少包括 3 个产品广告的图像。

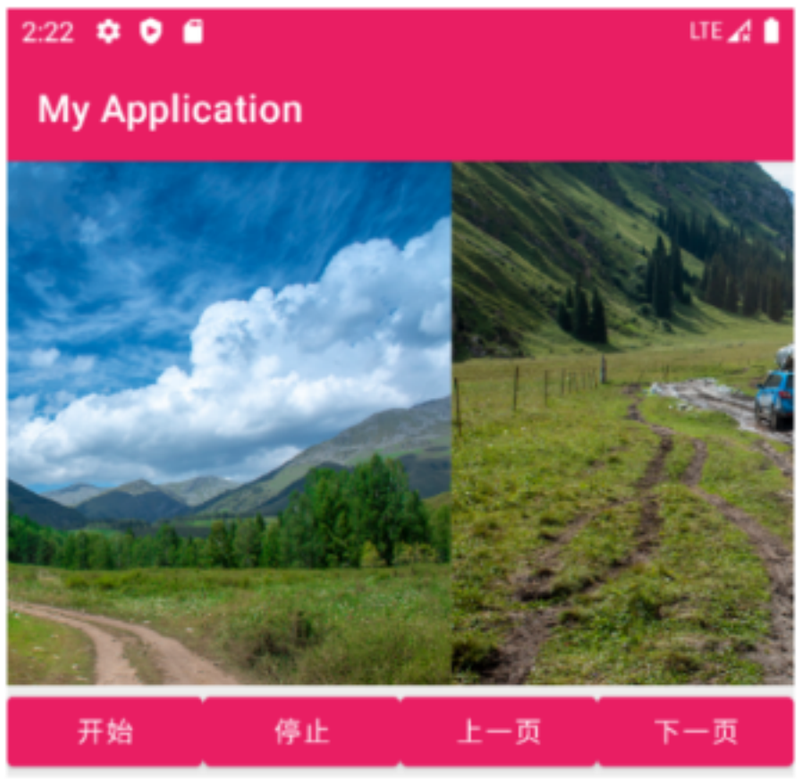
代码:

```
public class MainActivity extends AppCompatActivity {
    private ViewFlipper mVewFlipper;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        View view3 = View.inflate(MainActivity.this, R.layout.view_flipper_page3,
null);
        ImageView imageView4 = new ImageView(this);
        imageView4.setAdjustViewBounds(true);
        imageView4.setImageResource(R.mipmap.img_9546);
        mVewFlipper = findViewById(R.id.view_flipper);
        mVewFlipper.addView(imageView4);
        mVewFlipper.addView(view3,2);
        mVewFlipper.startFlipping();
        mVewFlipper.setOnClickListener(v -> Toast.makeText(MainActivity.this, "当
前显示的子视图索引号为"+mVewFlipper.getDisplayedChild(),
Toast.LENGTH_SHORT).show());
        Button startBtn = findViewById(R.id.button_start); //开始按钮
        startBtn.setOnClickListener(v -> mVewFlipper.startFlipping());
        Button stopBtn = findViewById(R.id.button_stop); //停止按钮
        stopBtn.setOnClickListener(v -> mVewFlipper.stopFlipping());
        Button previousBtn = findViewById(R.id.button_previous);
        previousBtn.setOnClickListener(v -> //上一页按钮
mVewFlipper.showPrevious());
        Button nextBtn = findViewById(R.id.button_next); //下一页按钮
        nextBtn.setOnClickListener(v -> mVewFlipper.showNext());
    }}
<?xml version="1.0" encoding="utf-8" ?>
<GridLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_gravity="top|fill"
    android:alignmentMode="alignMargins"
    android:background="#eeeeee"
    android:columnCount="4"
    android:rowCount="2">
    <ViewFlipper
        android:id="@+id/view_flipper"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:layout_columnSpan="4"
        android:flipInterval="3000"
        android:inAnimation="@anim/view_flipper_right_in"
        android:outAnimation="@anim/view_flipper_left_out">
        <include layout="@layout/view_flipper_page1" />
        <include layout="@layout/view_flipper_page2" />
    </ViewFlipper>
    <Button
        android:id="@+id/button_start"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1.0"
        android:text="开始" />
    <Button
        android:id="@+id/button_stop"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1.0"
        android:text="停止" />
    <Button
        android:id="@+id/button_previous"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1.0"
        android:text="上一页" />
    <Button
        android:id="@+id/button_next"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1.0"
        android:text="下一页" />
</GridLayout>
```

运行结果：


2:22
LTE
My Application
开始
停止
上一页
下一页


2:22
LTE
My Application
开始
停止
上一页
下一页

第五章：

p150 5.1 实现登录后显示用户名的效果。输入用户名、密码后登录，启动一个新的 Activity 显示用户名。

代码：

```
public class MainActivity extends AppCompatActivity {
    private static final int REQUEST_CODE = 101;
    private EditText mResultEditText;
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mResultEditText = findViewById(R.id.edit_text_result);
        findViewById(R.id.button_send).setOnClickListener(new
View.OnClickListener() {//发送数据（没有回传数据）
            public void onClick(View v) {
                Intent intent = new Intent(MainActivity.this, ReceiveActivity.class);
                intent.putExtra("id", 10407);
                intent.putExtra("msg", "MainActivity 发送的信息 1");
                startActivity(intent);
                mResultEditText.setText("");}});
findViewById(R.id.button_send_for_result).setOnClickListener(new
View.OnClickListener() {
            public void onClick(View view) {
                Intent intent = new Intent(MainActivity.this, ReceiveActivity.class);
                intent.putExtra("id", 20408);
                intent.putExtra("msg", "MainActivity 发送的信息 2");
                startActivityForResult(intent, REQUEST_CODE);
                mResultEditText.setText("");}}); }
    protected void onActivityResult(int requestCode, int resultCode, Intent data) {//
处理回调数据
        super.onActivityResult(requestCode, resultCode, data);
        mResultEditText.setText("正在处理回调数据");
        if (requestCode == REQUEST_CODE) { //判断请求码
            if (resultCode == ReceiveActivity.RESULT_CODE) {
                String result = data.getStringExtra("data");
                mResultEditText.setText(result);
            }else{
                mResultEditText.setText("没有回传数据");}}}}
public class ReceiveActivity extends AppCompatActivity {
    public static final int RESULT_CODE = 201;
    private EditText mIdEditText;
    private EditText mMsgEditText;
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_receive);
        Intent intent = getIntent();//获取传递数据
```

```
        int id = intent.getIntExtra("id", 0);
        String msg = intent.getStringExtra("msg");
        mIdEditText = findViewById(R.id.edit_text_id); //显示传递数据
        mIdEditText.setText(String.valueOf(id));
        mMsgEditText = findViewById(R.id.edit_text_name);
        mMsgEditText.setText(msg);
        findViewById(R.id.button_finish).setOnClickListener(view -> {
            finish();//关闭当前 Activity});
findViewById(R.id.button_finish_result).setOnClickListener(view -> {
    Intent intent1 = new Intent();//设置返回的数据//关闭（发送回传数据）
        intent1.putExtra("data", "已经查阅信息！ ");
        setResult(RESULT_CODE, intent1);
        finish();//关闭当前 Activity});}}
```

运行结果：


3:53
LTE
C0508
id
10407
msg
MainActivity发送的信息1
关闭（没有回传数据）
关闭（发送回传数据）


3:53
LTE
C0508
id
20408
msg
MainActivity发送的信息2
关闭（没有回传数据）
关闭（发送回传数据）

第六章：

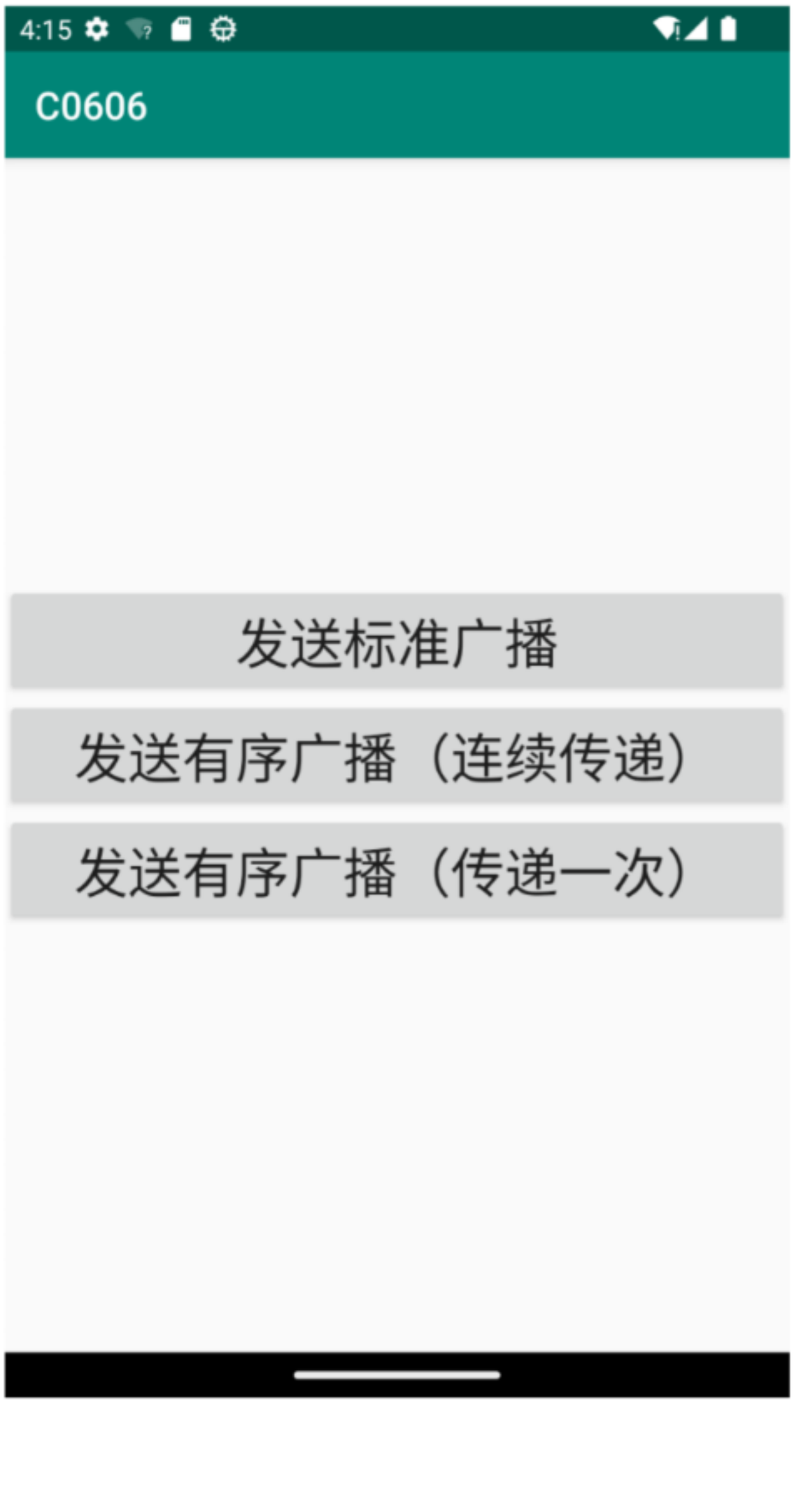
p176 6.3 通过广播接收器实现两个 App 之间通过自定义广播进行数据通信。

代码：

```
public class MainActivity extends AppCompatActivity {
        protected void onCreate(Bundle savedInstanceState) {
                super.onCreate(savedInstanceState);
                setContentView(R.layout.activity_main);
findViewById(R.id.button_send_broadcast).setOnClickListener(new
View.OnClickListener() {//发送标准广播
                public void onClick(View v) {
                        Intent intent=new Intent();
                        intent.setAction("MyBroadcastReceiver.Custom");
                        intent.putExtra("info","悄悄地告诉你一个秘密^6^");
                        intent.addFlags(Intent.FLAG_ACTIVITY_PREVIOUS_IS_TOP);
                        sendBroadcast(intent);    }});
findViewById(R.id.button_send_ordered_broadcast).setOnClickListener(new
View.OnClickListener() {//发送有序广播（连续传递）
                public void onClick(View v) {
                        Intent intent=new Intent();
                        intent.setAction("OrderedBroadcast.Custom");
                        intent.putExtra("info","悄悄地告诉你一个秘密^6^");
                        intent.addFlags(Intent.FLAG_ACTIVITY_PREVIOUS_IS_TOP);
                        sendOrderedBroadcast(intent,null);}});
findViewById(R.id.button_send_ordered_broadcast_stop).setOnClickListener(new
View.OnClickListener() {//发送有序广播（传递一次）
                public void onClick(View v) {
                        Intent intent=new Intent();
                        intent.setAction("OrderedBroadcast.Custom");
                        intent.putExtra("stop",true);
                intent.putExtra("info","悄悄地告诉你一个秘密  不要告诉别人哦^6^");
intent.addFlags(Intent.FLAG_ACTIVITY_PREVIOUS_IS_TOP);
                        sendOrderedBroadcast(intent,null);}});}}
public class MyBroadcastReceivers1 extends BroadcastReceiver {
        private static final String TAG = "MyBroadcastReceiver";
        public void onReceive(Context context, Intent intent) {
                Log.d(TAG, "----------Receivers1----------");
                Log.d(TAG, "Action: " + intent.getAction());
                Log.d(TAG, "URI: " + intent.toUri(Intent.URI_INTENT_SCHEME));
                Log.d(TAG, "自定义广播的 info：" + intent.getStringExtra("info"));}}
public class MyBroadcastReceivers2 extends BroadcastReceiver {
        private static final String TAG = "MyBroadcastReceiver";
        public void onReceive(Context context, Intent intent) {
                Log.d(TAG, "----------Receivers2----------");
                Log.d(TAG, "Action: " + intent.getAction());
```

```
        Log.d(TAG, "URI: " + intent.toUri(Intent.URI_INTENT_SCHEME));
        Log.d(TAG, "自定义广播的 info：" +intent.getStringExtra("info"));
        if(intent.getBooleanExtra("stop",false)){
            abortBroadcast();}}}
```

运行结果：




4:17
C0606
发送标准广播
发送有序广播（连续传递）
发送有序广播（传递一次）

第七章：

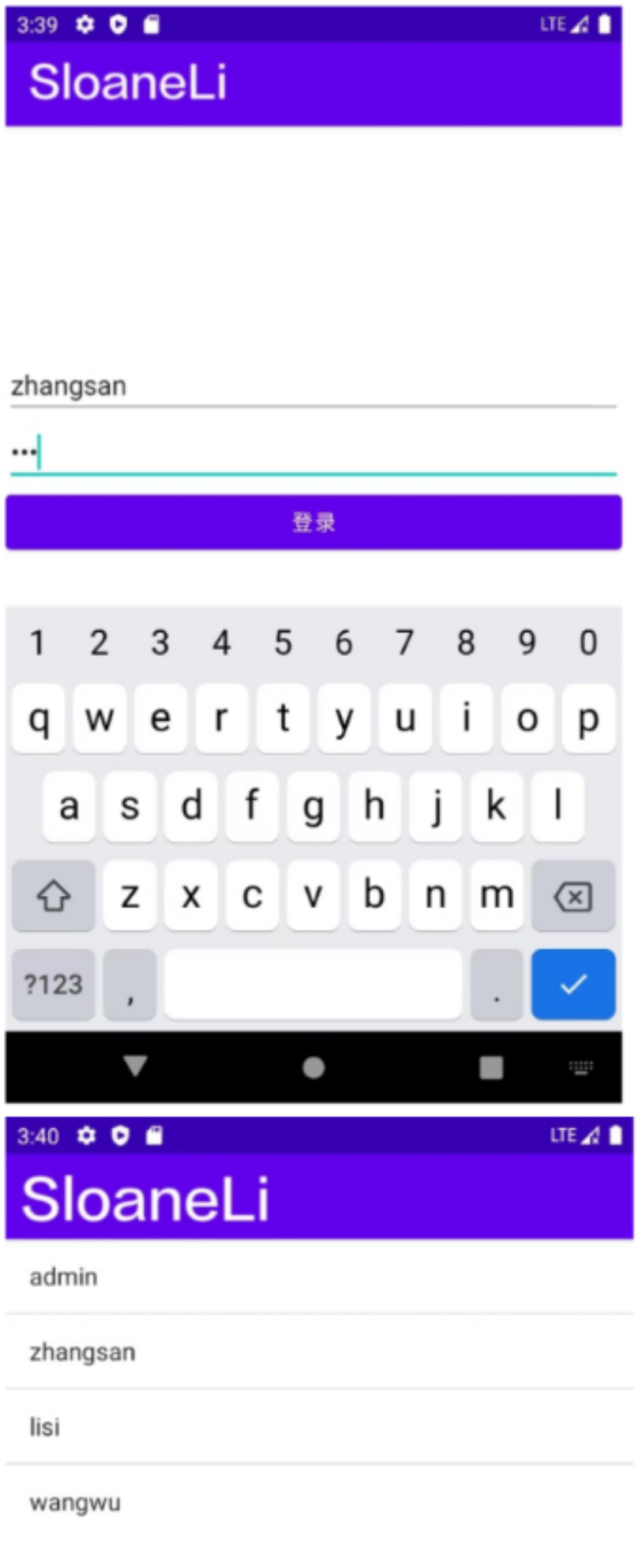
p213 7.2 使用 SQLite 实现记录登录时间的功能，并且能够查询登录的历史记录。

代码：

```
public class SQLiteDBHelper extends SQLiteOpenHelper {
      static final String CREATE_SQL[] = {//  创建数据库
                  "CREATE TABLE news (" +
"_id    INTEGER NOT NULL PRIMARY KEY AUTOINCREMENT," +"title    varchar," +"content    varchar," +"keyword    varchar," +"category    varchar," +"author    INTEGER," +"publish_time    varchar" +")",
                  "CREATE TABLE user (" +"_id    INTEGER NOT NULL PRIMARY KEY AUTOINCREMENT," +"name    varchar," +"password    varchar," +"email    varchar," +"phone    varchar," +"address    varcher" +")",
   "INSERT INTO userVALUES(1,'admin',123,'admin@163.com',123,'沈阳')",
   "INSERT INTO user VALUES (2,'zhangsan',123,'zhangsan@163.com',123,'北京')",
   "INSERT INTO user VALUES (3,'lisi',123,'lisi@163.com',123,'上海')",
  "INSERT INTO user VALUES (4,'wangwu',123,'wangwu@163.com',123,'深圳')"};
      public SQLiteDBHelper(@Nullable Context context, @Nullable String name, int version) {//  调用父类的构造方法(便于之后进行初始化赋值)
            super(context, name, null, version);    }
      public void onCreate(SQLiteDatabase db) {
            Log.i("sqlite_____", "create Database");//  创建数据库
            for (int i = 0; i < CREATE_SQL.length; i++) {//  执行  SQL  语句
                  db.execSQL(CREATE_SQL[i]); }
            Log.i("sqlite_____", "Finished Database");//  完成数据库的创建}
public class SQLiteListActivity extends AppCompatActivity {
      ListView listView01;
      SQLiteDBHelper dbHelper;
      protected void onCreate(Bundle savedInstanceState) {
            super.onCreate(savedInstanceState);
            setContentView(R.layout.activity_sqlite_list);
            Toast.makeText(this,"登录成功！ ",Toast.LENGTH_LONG).show();
            listView01 = findViewById(R.id.listView01);
            dbHelper = new SQLiteDBHelper(this,"sqlite.db",1);
            SQLiteDatabase db = dbHelper.getReadableDatabase();
            Cursor cursor = db.query("user",new
String[]{"_id","name"},null,null,null,null,null,null);
            SimpleCursorAdapteradapter=newSimpleCursorAdapter(this,android.R.layout.simple_list_item_1,cursor,new String[]{"_id","name"},new int[]{android.R.id.text1,android.R.id.text1},0);
            listView01.setAdapter(adapter);}
      protected void onDestroy() {
            super.onDestroy();
            dbHelper.close();}}
public class SQLiteLoginActivity extends AppCompatActivity {
```

```
SQLiteDBHelper dbHelper;
EditText etUsername;
EditText etPassword;
protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_sqlite_login);
        dbHelper = new SQLiteDBHelper(this,"sqlite.db",1);
        etUsername = findViewById(R.id.etUsername);
        etPassword = findViewById(R.id.etPassword);}
    public void onClick(View view) {
            String username = etUsername.getText().toString();
            String password = etPassword.getText().toString();
            SQLiteDatabase db = dbHelper.getReadableDatabase();
            Cursor cursor = db.query("user",new String[]{"name","password"},
"name=?",new String[]{username},null,null,null,"0,1");
if(cursor.moveToNext()) {@SuppressLint("Range") String dbPassword =
cursor.getString(cursor.getColumnIndex("password"));
            cursor.close();
                if(password.equals(dbPassword)) {
                    Intent intent = new Intent(this, SQLiteListActivity.class);
                    startActivity(intent); //  启动
                    return;}}
        cursor.close();//  跳转失败也要进行关闭
        Toast.makeText(this,"输入信息有误！",Toast.LENGTH_LONG).show();}
public void onDestroy() {
        super.onDestroy();
        dbHelper.close();}}
```

运行结果：


3:39
LTE
SloaneLi
zhangsan
登录
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123
3:40
LTE
SloaneLi
admin
zhangsan
lisi
wangwu

第八章：

p271 8.1 使用 Camera2 实现定时拍照的功能。

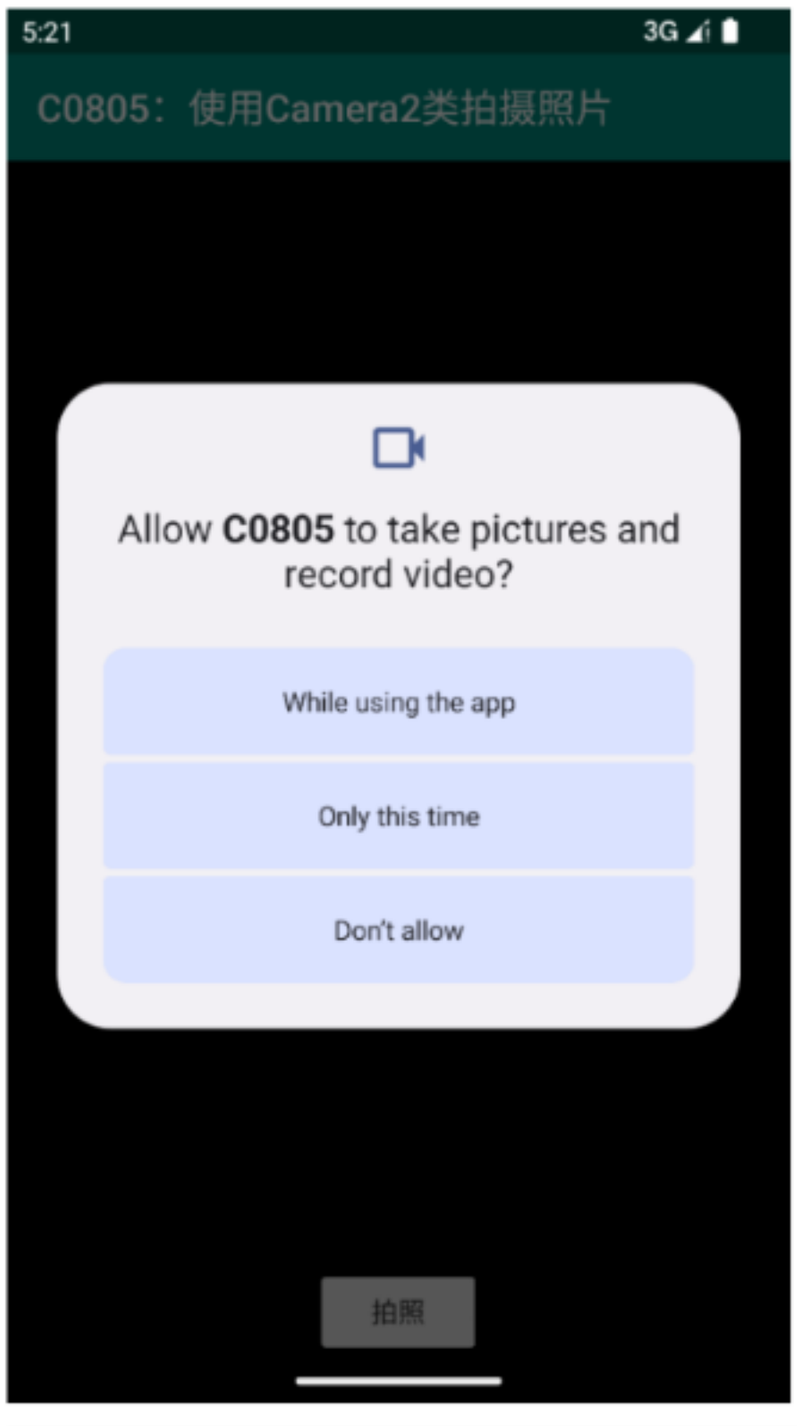
代码：

```
 public void onClick(View view) {
            takePhoto();
    }
    @SuppressLint("MissingPermission") //  打开摄像头
    private void openCamera(int width, int height) {
        Log.d(TAG, "openCamera()");
        if (!Permissions.hasPermissionsGranted(mContext, PERMISSIONS))
{Permissions.requestPermissions(mContext, PERMISSIONS);
                return;}
            try {// 2500 毫秒内请求获取 1 个许可，否则抛出异常。
                if (!mSemaphore.tryAcquire(2500, TimeUnit.MILLISECONDS)) {
                    throw new RuntimeException("打开摄像头超时");}
                mCameraManager = (CameraManager)
getSystemService(Context.CAMERA_SERVICE);
            CameraCharacteristics characteristics = //  获取摄像头支持的属性特征
mCameraManager.getCameraCharacteristics(mCameraId);
             StreamConfigurationMap map =
characteristics.get(CameraCharacteristics.SCALER_STREAM_CONFIGURATION_MAP)
; //  获取摄像头捕捉的最大尺寸读取图片
                                Size largest =
Collections.max(Arrays.asList(map.getOutputSizes(ImageFormat.JPEG)), new
Util.CompareSizesByArea());
                mImageReader = ImageReader.newInstance(largest.getWidth(),
largest.getHeight(), ImageFormat.JPEG, 2); //  选择最佳预览尺寸
                mPreviewSize = Util.chooseOptimalSize(mContext,
map.getOutputSizes(SurfaceTexture.class), width, height, MAX_PREVIEW_WIDTH,
MAX_PREVIEW_HEIGHT, largest);
                mPreviewView.setAspectRatio(mPreviewSize.getHeight(),
mPreviewSize.getWidth());//  设置预览尺寸
                mCameraManager.openCamera(mCameraId, //打开摄像头
mCameraDeviceStateCallback, mBackgroundHandler);
            } catch (CameraAccessException e) {
                e.printStackTrace();
            } catch (InterruptedException e) {
                throw new RuntimeException("打开摄像头被中断", e);}}
        private void startPreview() { //  创建摄像头预览会话
            Log.d(TAG, "startPreview()");
            if (null == mCameraDevice || !mPreviewView.isAvailable() || null ==
mPreviewSize) { return; }
            try {
                closePreview();//  创建摄像头预览会话
                SurfaceTexture texture = mPreviewView.getSurfaceTexture();
```

```
                texture.setDefaultBufferSize(mPreviewSize.getWidth(),
mPreviewSize.getHeight());//  设置预览输出的 Surface
                Surface surface = new Surface(texture);
                mCaptureRequestBuilder =
mCameraDevice.createCaptureRequest(CameraDevice.TEMPLATE_PREVIEW);
                mCaptureRequestBuilder.addTarget(surface);
mCameraDevice.createCaptureSession(Arrays.asList(surface,
mImageReader.getSurface()), new CameraCaptureSession.StateCallback() {
public void onConfigured(CameraCaptureSession cameraCaptureSession) {
                        mCaptureSession = cameraCaptureSession;
                        updatePreview();}
                    public void onConfigureFailed(CameraCaptureSession
cameraCaptureSession) {
                        Toast.makeText(mContext, "摄像头配置失败",
Toast.LENGTH_LONG).show();}}, mBackgroundHandler);
            } catch (CameraAccessException e) {
                e.printStackTrace();}}
    private void takePhoto() {       //  拍照
        Log.d(TAG, "takePhoto()");
        if (null == mCameraDevice || !mPreviewView.isAvailable() || null ==
mPreviewSize) {
            return;}
        try {
            closePreview();
            mCaptureRequestBuilder =
mCameraDevice.createCaptureRequest(CameraDevice.TEMPLATE_STILL_CAPTURE);
// 设置预览的缓冲区大小
            SurfaceTexture texture = mPreviewView.getSurfaceTexture();
            texture.setDefaultBufferSize(mPreviewSize.getWidth(),
mPreviewSize.getHeight());// 设置预览的 Surface
            List<Surface> surfaces = new ArrayList<>();
            Surface previewSurface = new Surface(texture);
            surfaces.add(previewSurface);
            mCaptureRequestBuilder.addTarget(previewSurface);
            Surface imageReaderSurface = mImageReader.getSurface();
            surfaces.add(imageReaderSurface);
            mCaptureRequestBuilder.addTarget(imageReaderSurface);
            mCameraDevice.createCaptureSession(surfaces, new
CameraCaptureSession.StateCallback() {
                public void onConfigured(@NonNull CameraCaptureSession
cameraCaptureSession) {
                    Log.d(TAG, "CameraCaptureSession.onConfigured()");
                    mCaptureSession = cameraCaptureSession;
mImageReader.setOnImageAvailableListener(mOnImageAvailableListener,
```

```
mBackgroundHandler);
                            mCaptureRequest = mCaptureRequestBuilder.build();
                            try {//  拍摄照片
                           mCaptureSession.capture(mCaptureRequest, null, null);
                            } catch (CameraAccessException e) {
                                e.printStackTrace();}}
                        public void onConfigureFailed(CameraCaptureSession
cameraCaptureSession) {
                            Toast.makeText(mContext, "摄像头配置失败",
Toast.LENGTH_LONG).show();}}, mBackgroundHandler);
                } catch (CameraAccessException e) {
                    e.printStackTrace();}}
        public static void requestPermissions(final Context context, String[]
permissions) {
                if (shouldShowRequestPermissionRationale(context, permissions)) {
                    String msg = "";
                    for (int i = 0; i < permissions.length; i++) {
                        if (i == 0) {
                            msg = permissions[i];
                        } else {
                            msg = msg + "\n" + permissions[i];}}
```

运行结果：


5:21
3G
C0805：使用Camera2类拍摄照片
Allow C0805 to take pictures and record video?
While using the app
Only this time
Don't allow
拍照

5:21
3G
C0805：使用Camera2类拍摄照片
Allow C0805 to access photos, videos, music, and audio on this device?
Allow
Don't allow
拍照


5:22
3G
C0805：使用Camera2类拍摄照片
拍照


5:23
3G
照片预览